

CS471 – Web Technologies

Lab 1 – The Internet Protocols

Student Name	Lamis Aloraini
Student ID	431201162

Part 1: Capturing HTTP Traffic

The captured packets show HTTP GET requests and server responses, including 301 Moved Permanently and 200 OK.

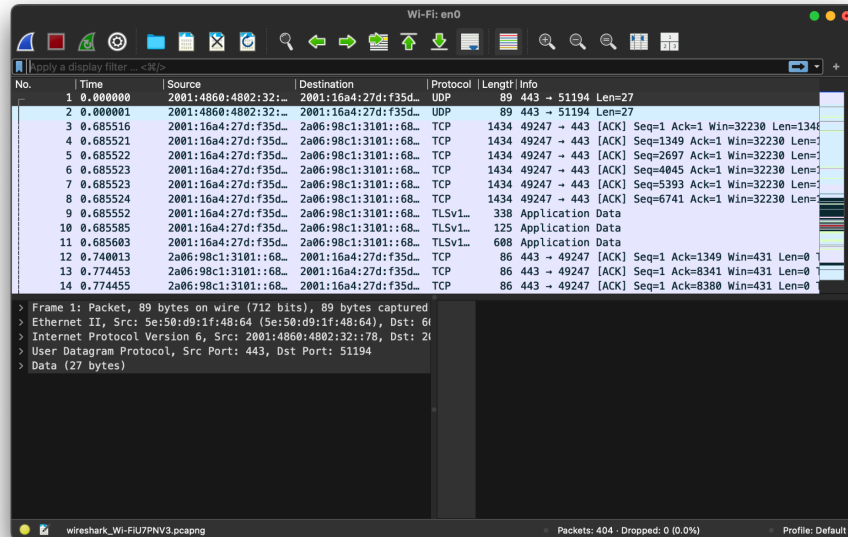


Figure 1. Wireshark capturing traffic on the Wi-Fi interface.

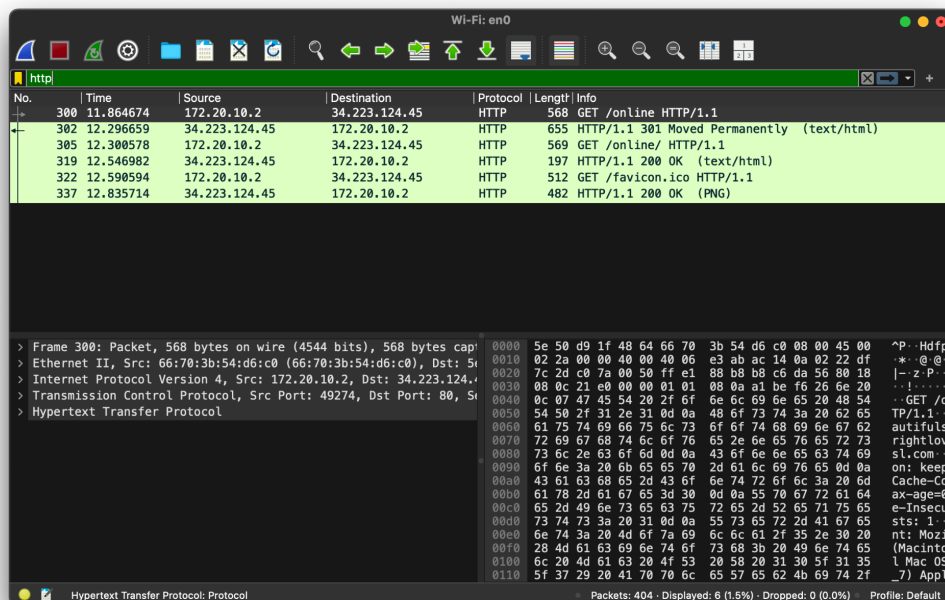


Figure 2. HTTP traffic filtered in Wireshark.

Part 2: Analyzing TCP/IP Traffic

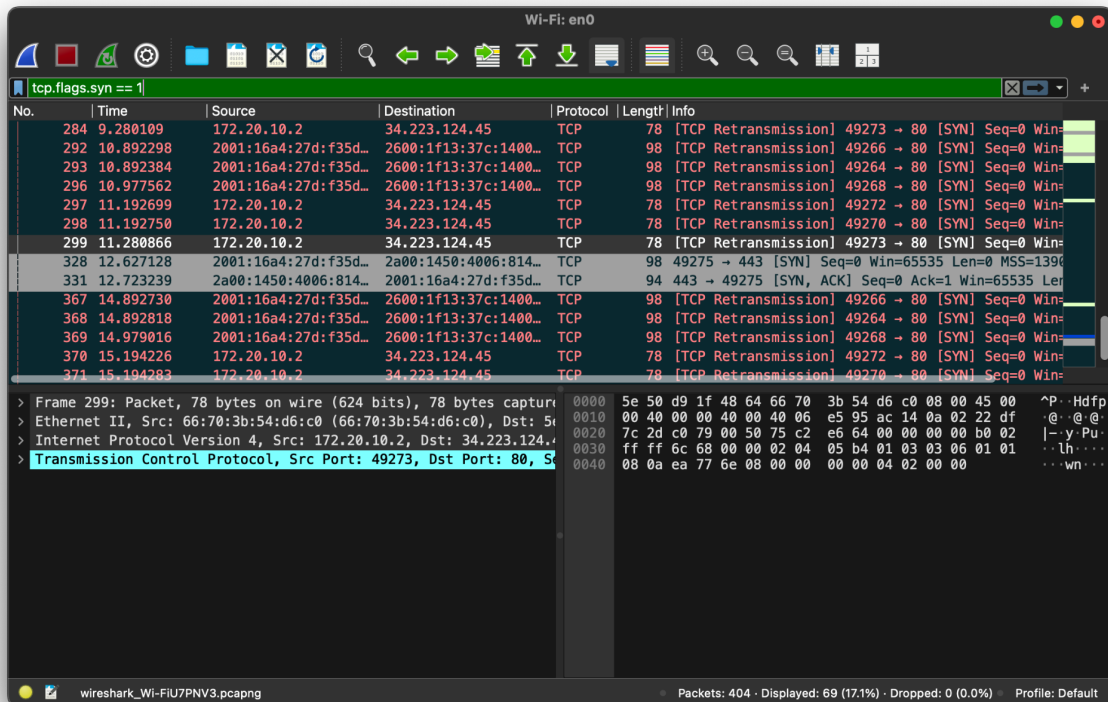


Figure 3. SYN packets used to inspect TCP connection establishment.

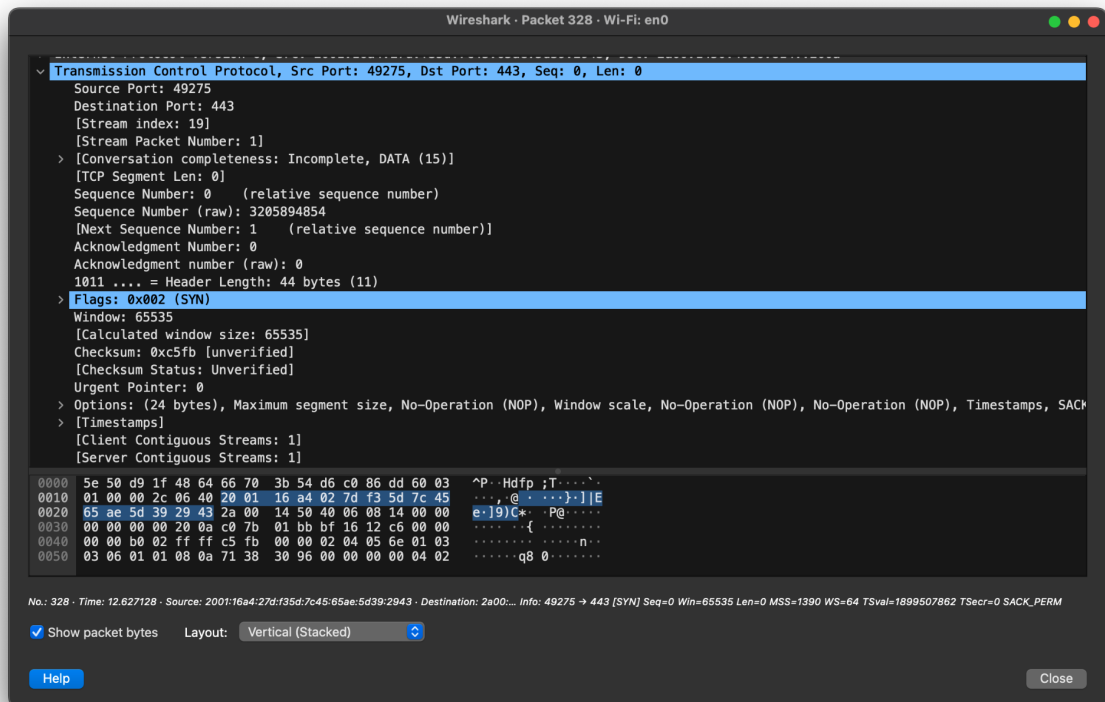


Figure 4. TCP SYN packet details showing sequence and acknowledgment information.

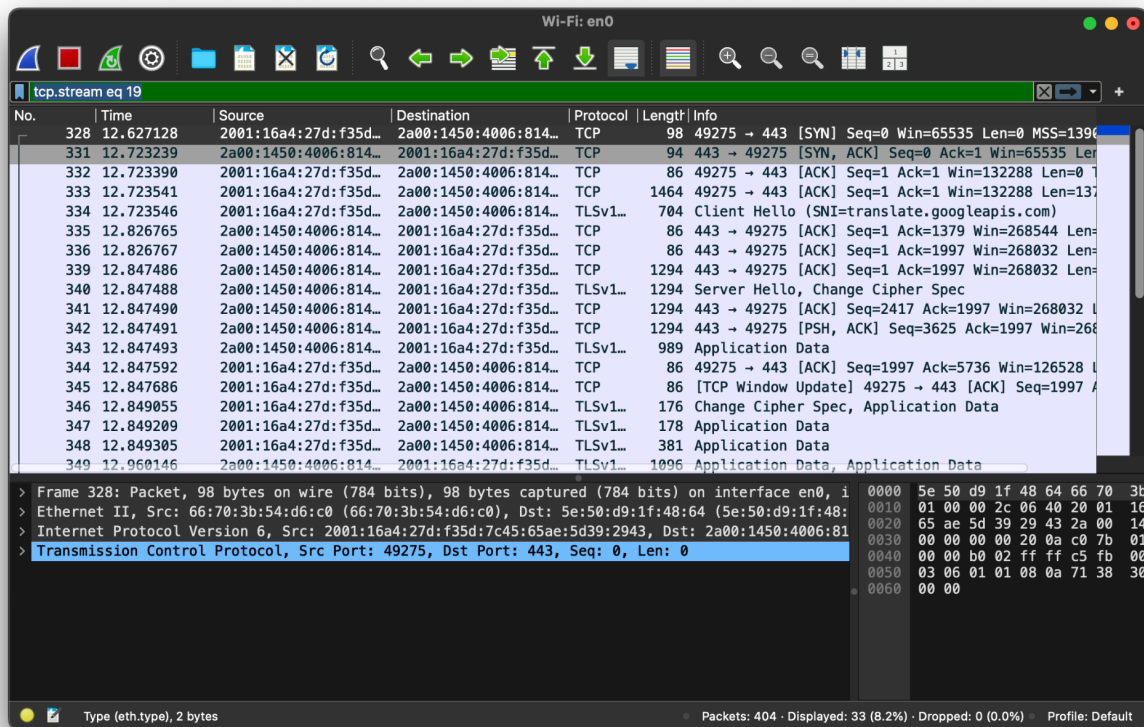


Figure 5. TCP three-way handshake (SYN, SYN-ACK, ACK) followed by data transfer.

A TCP packet associated with the HTTP connection on port 80 was selected and Follow → TCP Stream was used to display the client-server conversation.

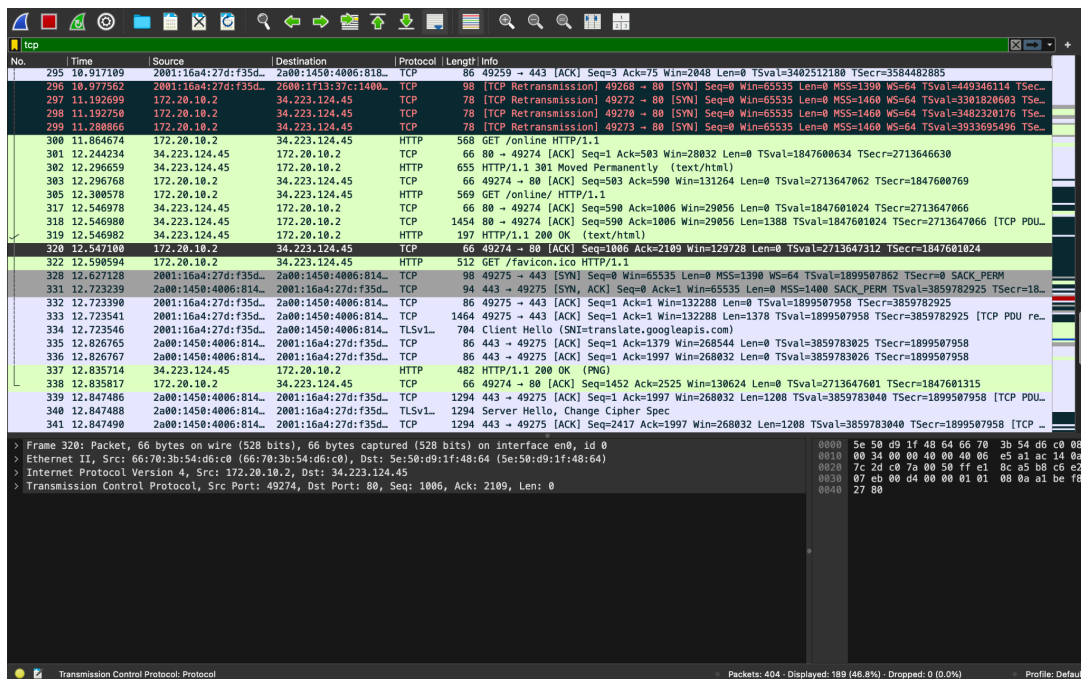


Figure 6. TCP packets associated with the HTTP connection on port 80.

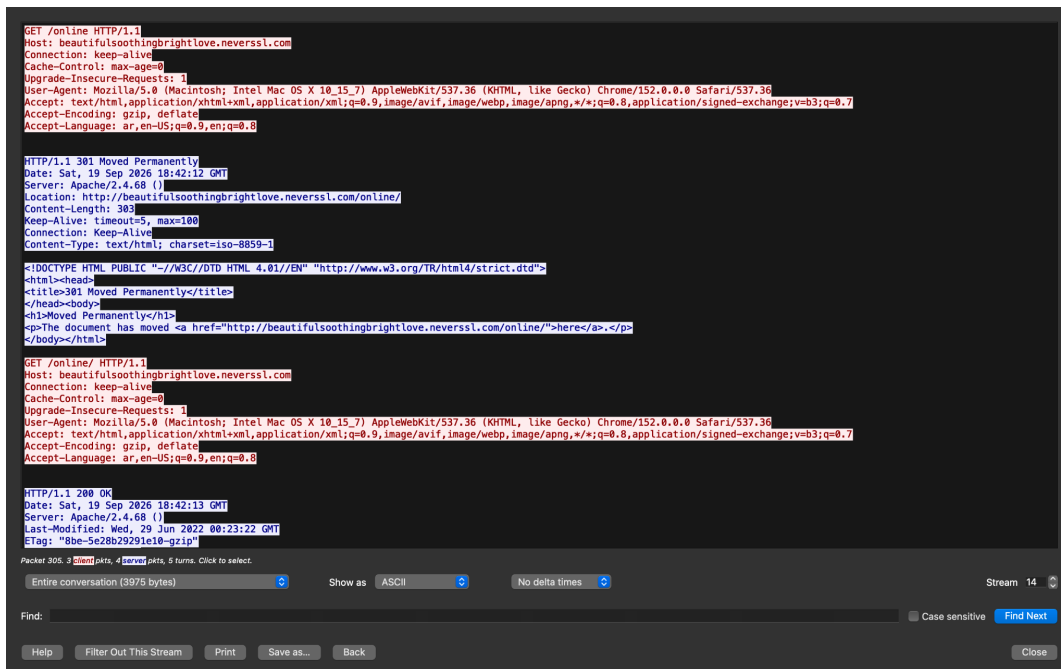


Figure 7. Follow TCP Stream showing HTTP GET requests and server responses.

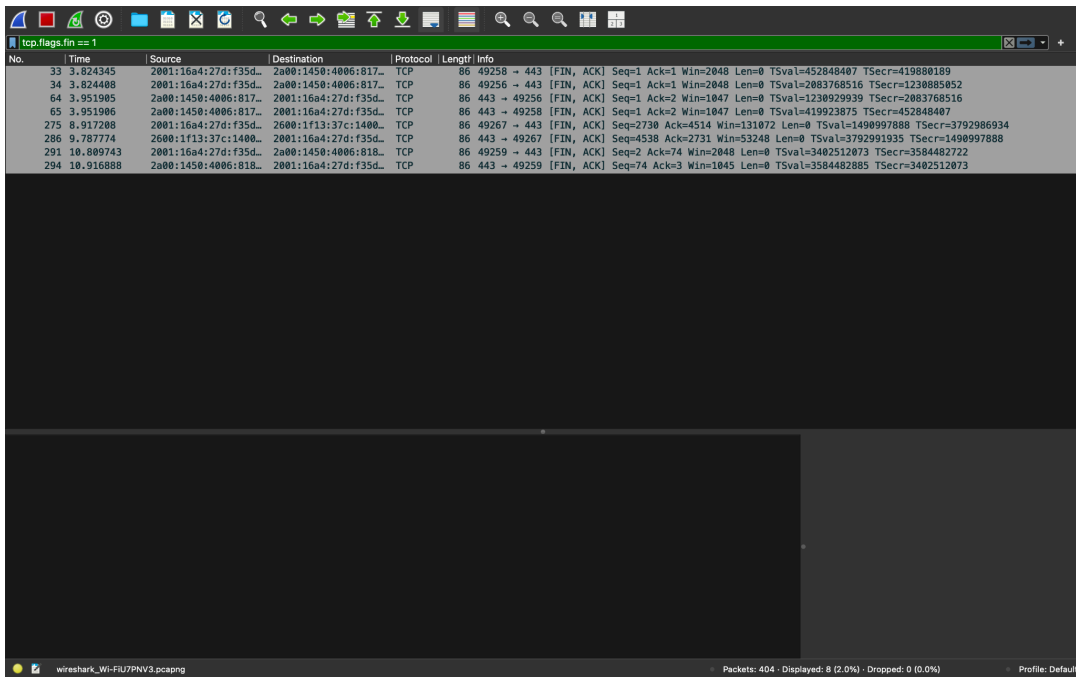


Figure 8. TCP termination showing FIN/ACK packets.

Part 3: Capturing and Analyzing UDP Traffic

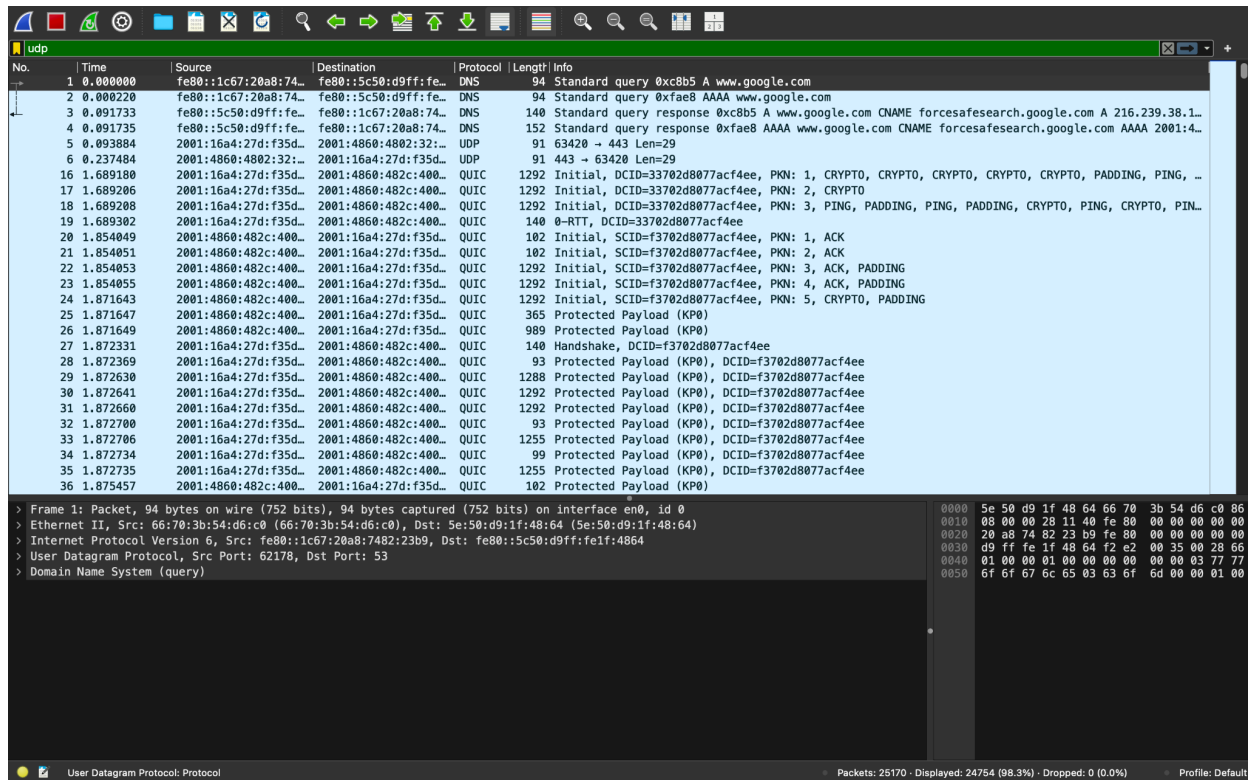


Figure 9. UDP-based traffic generated and captured while the network application was running.

A UDP packet was selected and its header was expanded. The inspected packet shows Source Port 63420, Destination Port 443, UDP Length 37, and a UDP payload/Data size of 29 bytes.

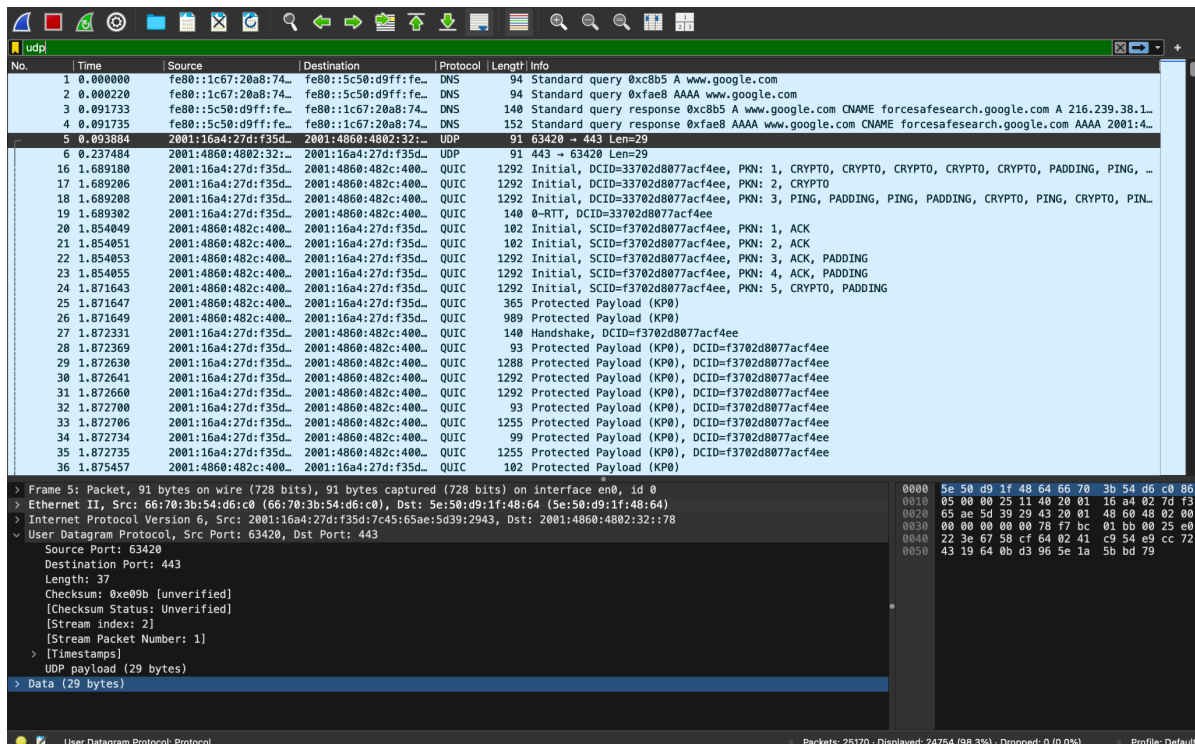


Figure 10. UDP packet details showing source port, destination port, length, payload, and data.

Part 4: Comparing TCP and UDP

Task 1: Fill in the following table and provide reasons.

	TCP or UDP	Reasons
Reliability and Connection Establishment	TCP	TCP establishes a connection before sending data and provides reliable delivery using acknowledgments and retransmissions.
Data Integrity and Ordering	TCP	TCP ensures that packets are delivered correctly and in the proper order.

Task 2: Identify the use Cases and Performance of TCP and UDP.

	TCP	UDP
Use cases	Web browsing, email, and file transfer	Streaming, VoIP, and online gaming
Performance	More reliable but has more overhead	Faster with lower overhead, but less reliable